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Graded Assignment • 1h

Due May 18, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

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1. **Create a prompt for a large language model, such as ChatGPT, to describe how large language models work. Ask at least three follow-up questions based on the output. Provide the prompts and outputs from the conversation as the answer.**

1 / 1 point

Explain how Large Language Models (LLMs) like yourself work under the hood. Break down the core concepts such as training phases, neural networks, and how you choose the next word in a sequence. Keep the explanation accessible but technical enough to understand the foundational mechanics.
Initial Model Output:
At their core, Large Language Models (LLMs) operate as highly sophisticated statistical prediction engines. They do not "think" or understand concepts the way humans do; instead, they compute probabilities based on patterns found in massive datasets. Here is how they work under the hood:

The Architecture (Transformers): Modern LLMs are built on a deep learning architecture called the Transformer. This architecture uses an "attention mechanism" that allows the model to look at a whole sentence at once and compute how different words relate to one another, regardless of how far apart they are in the text.

Data Representation (Tokens and Embeddings): Before processing, text is broken down into smaller pieces called tokens (which can be whole words or syllables). These tokens are translated into long lists of numbers called vectors or "embeddings." Words with similar meanings or contexts are mapped close together in a high-dimensional mathematical space.

The Pre-training Phase: This is the most computationally expensive step. The model is fed billions of pages of text from the internet, books, and articles. It plays a game of "fill-in-the-blank," guessing the hidden next word in a sentence over and over. By adjusting billions of internal algebraic dials (called parameters) through trial and error, it learns grammar, facts, and reasoning patterns.

The Fine-Tuning Phase (RLHF): To make the model safe and helpful, it undergoes Reinforcement Learning from Human Feedback (RLHF). Human trainers grade the model's responses, steering it away from harmful behaviors and teaching it to follow explicit conversational instructions.

Next-Token Prediction: When you type a prompt, the model converts your input into embeddings, processes it through its neural network parameters, and outputs a probability distribution for the next possible word. It selects the highest-probability or most contextually relevant word, appends it to your prompt, and repeats the entire loop to choose the word after that.

Follow-up Prompt (Asking 3 Questions at Once):
Plaintext
Based on your explanation, please answer the following three follow-up questions:
1. What exactly are 'parameters,' and how do they change during the pre-training phase?
2. If you are predicting the next token based on probabilities, why don't you give the exact same answer every time the same prompt is entered?
3. How does the attention mechanism help you understand context better than older language models?
Follow-up Model Output:
Here are the answers to your three follow-up questions:

What parameters are and how they change: Parameters are the internal mathematical weights or configurations within the neural network. Think of them as billions of tiny volume dials. During pre-training, when the model makes an incorrect next-word guess, an optimization algorithm calculates how far off the guess was and uses a mathematical process called backpropagation to slightly turn these dials. Over billions of iterations, these dials land on configurations that successfully map language patterns.

Why outputs vary (Temperature and Top-P): If an LLM always picked the absolute highest-probability word, its responses would be repetitive and robotic. To prevent this, developers use parameters like "Temperature." When temperature is set above zero, the model randomly samples from a pool of top-scoring words instead of just choosing the single top choice. This introduces dynamic creativity and variation across identical prompts.

How the attention mechanism improves context: Older language models processed words sequentially (one by one, left to right). By the time they reached the end of a long paragraph, they would mathematically "forget" the context at the beginning. The Transformer's self-attention mechanism assigns mathematical weights to every single word in a prompt simultaneously, mapping how much attention the model should pay to each word relative to the others, vastly extending its contextual memory.

2. **Create a prompt for a large language model, such as ChatGPT, that asks it to perform an instruction other than answering a question. For example, ask ChatGPT to create a meal plan, plan a trip, create a recipe, or help you diagnose a problem. Provide the prompts and outputs from the conversation as the answer.**

1 / 1 point

Act as an expert travel guide. Create a detailed 3-day weekend itinerary for a trip to Kyoto, Japan, optimized for a first-time visitor. Focus on historical temples, traditional dining experiences, and efficient geographic routing so that I am not spending too much time traveling between sites. Format the output with clear daily schedules.
Here is an optimized 3-day cultural itinerary for your first visit to Kyoto, designed to minimize transit times by grouping attractions geographically:

Day 1: Eastern Kyoto (Higashiyama District)

08:30 AM – Kiyomizu-dera Temple: Start early to beat the crowds at this iconic wooden temple built over a hillside, offering panoramic views of the city.

10:30 AM – Historic Streets: Walk down the preserved stone-paved streets of Ninenzaka and Sannenzaka, stopping by traditional wooden machiya tea houses.

12:30 PM – Lunch: Try local Kyoto-style udon or a tofu-based set meal in the Higashiyama neighborhood.

02:00 PM – Yasaka Shrine & Maruyama Park: Explore the vibrant vermilion gates of the shrine and relax in the adjoining park.

04:00 PM – Gion District Walk: Stroll through the historic entertainment district where you might catch a glimpse of a geisha heading to an evening appointment.

06:30 PM – Dinner: Enjoy a traditional multi-course Kaiseki dinner along the scenic Kamogawa River.

Day 2: Western Kyoto (Arashiyama & The Golden Pavilion)

08:00 AM – Arashiyama Bamboo Grove: Arrive early to experience the towering stalks of bamboo in peace. Walk through the adjacent Tenryu-ji Temple Zen gardens.

11:00 AM – Togetsukyo Bridge: Take a scenic walk along the Katsura River and pick up local street snacks.

12:30 PM – Lunch: Enjoy a traditional Buddhist vegetarian meal (shojin ryori) inside a temple sub-compound.

02:30 PM – Kinkaku-ji (The Golden Pavilion): Take a short taxi or bus ride north to view this spectacular Zen temple, whose top two floors are completely covered in gold leaf.

06:00 PM – Dinner: Head to downtown Kyoto (Pontocho Alley) for a yakitori or sushi dinner in a narrow, atmospheric alleyway.

Day 3: Southern Gateways & Departure

07:30 AM – Fushimi Inari Shrine: Hike up the mountain path lined with thousands of vibrant red Torii gates. Walking the lower loops takes about an hour and provides incredible photo opportunities.

11:00 AM – Tofuku-ji Temple: Located just one train stop away, this temple features some of the best Zen rock gardens and valley bridge views in Kyoto with far fewer crowds.

01:00 PM – Lunch: Grab a quick, delicious bowl of ramen or hand-dipped soba near Kyoto Station before preparing for your departure.